Counting Secondary Care: Multitasking, Measured Unpaid Work and the Gender Gap in India

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Abstract

Standard measures of unpaid work, including SDG indicator 5.4.1, count only the main activity in each time slot, so care done alongside other tasks is invisible. India's Time Use Survey 2024 records every activity of ten minutes or more in each half-hour slot and asks which one was the main activity. Using all 10,211,478 activity records, I measure secondary activity for 388,847 diarists aged 15 and above. Women record 216 minutes of secondary activity a day and men 177. Once counted, secondary childcare is 22% of women's childcare, and the women-to-men childcare ratio moves from 3.5 to 3.6. Because the survey does not document how duration is assigned to secondary records, I re-estimate the inclusive measure under four duration conventions; the widening of the gender gap survives all of them. The gender gap in secondary childcare is larger in households with a child under six, and it is larger among the employed than among those not employed. Women are more likely than men to have free time shared with care or domestic work. Under a replacement-cost valuation with a single wage, counting secondary childcare raises the imputed value of women's childcare by between 5% and 28%, whatever wage is used. Primary-activity statistics therefore understate both the volume of women's unpaid work and the gender gap in it.

Keywords: multitasking, secondary activities, childcare, time intensity, leisure quality, SDG 5.4.1, unpaid work, India

JEL codes: D13, J16, J22, I31

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1 Introduction

Time-use statistics rest on a simple accounting rule: a day has 1,440 minutes, and each minute is assigned to one activity. The rule makes the statistics additive and comparable, and it underlies the official indicator for Sustainable Development Goal target 5.4, the proportion of time spent on unpaid domestic and care work (indicator 5.4.1). But much of what households do does not fit the rule. A woman who cooks while watching a toddler, feeds a child while eating, or listens for an elderly parent while resting is doing two things at once. If only the "main" activity is counted, one of them disappears. Feminist economists have long argued that the disappearing activity is disproportionately women's care work, and that the rule therefore understates both women's work and the intensity of their days (Floro 1995; Floro and Miles 2003; Folbre 2006). Sociologists of time have made the matching argument about leisure: equal minutes of free time are not equal leisure if one person's free time is interrupted by, or shared with, the care of others (Bittman and Wajcman 2000; Mattingly and Bianchi 2003; Sullivan 1997).

The point matters for three literatures. For the measurement of childcare, estimates of parents' time with children vary severalfold depending on whether secondary and supervisory care are counted (Zick and Bryant 1996; Folbre et al. 2005; Allard et al. 2007). For gender inequality, a measure that counts only primary activities may understate the gender gap in unpaid work if women do more of their care alongside other tasks, or overstate it if men's care is the more often secondary. And for the valuation of unpaid work, which multiplies hours by a wage, the hours that enter the calculation determine the answer (Goldschmidt-Clermont 1982; Abraham and Mackie 2005; Landefeld et al. 2009). Most of the evidence on these questions comes from Australia, Western Europe and the United States, where diaries record a secondary activity column or, as in the American Time Use Survey, ask a summary question about children "in your care" (Allard et al. 2007). Systematic evidence for low- and middle-income countries, where women's paid, own-use and care work are often performed in and around the home at the same time, is scarce (Esquivel et al. 2008; Hirway and Jose 2011).

This paper uses the India Time Use Survey 2024 (National Statistics Office 2025) to measure how much of women's and men's activity is secondary, how much of that is care, and what counting it does to the measured gender gap in childcare and unpaid work. The survey has three features that make it well suited to the task. It records, in each of 48 half-hour slots in the diary day, every activity that lasted ten minutes or more. It asks the respondent whether more than one activity was performed in the slot and, if so, whether any were simultaneous. And it asks which activity was the "most important" in the slot, flagging it as the major activity and the others as minor. Because every record carries a start time, each minor record can be matched to the major episode during which it occurred, so it is possible to see not only how much care is secondary but what the carer was doing at the time.

I build person-level measures from all 10,211,478 activity records and analyse 388,847 diarists

aged 15 and above: secondary minutes in total and by category; the number of secondary records; minutes of major activity in slots flagged as containing multiple activities; secondary childcare and an inclusive childcare measure; an inclusive version of the SDG 5.4.1 numerator; the share of eating recorded as secondary; and indicators of free time shared with care or domestic work and of meals eaten while working.

The main results are as follows. Women record 39.0 minutes (SE 1.1) more secondary activity a day than men, and women are 3.3 percentage points (SE 0.2) more likely to record any. Women's secondary childcare averages 14 minutes a day, against 3 minutes for men, and makes up 22% of their childcare once it is counted (20% for men). Counting it widens the measured gender gap in childcare: the women-to-men ratio moves from 3.5 on the primary-activity measure to 3.6 on the inclusive measure. The conditional gap in secondary childcare, net of age, education, marital status, household composition, employment, consumption, state and day of the week, is 7.4 minutes (SE 0.2), and the gap between women and men living in the same household is 7.1 minutes (SE 0.2).

Two parts of the analysis address the main measurement problems. First, the survey documentation available to me does not state how the release assigns duration to minor records. Rather than treat recorded duration as given, I re-estimate the inclusive childcare measure under four conventions: recorded duration, half of recorded duration, a floor of ten minutes per record (the survey's recording threshold), and recorded duration net of any secondary childcare that overlaps primary childcare. The gender gap widens under every convention, and the inclusive ratio lies between 3.6 and 3.6. Second, I ask who does the hidden care. The gender gap in secondary childcare is larger in households with a child under six, and larger among the employed than among those not employed.

I then use the results to quantify, without inventing a wage, what single-activity accounting implies for valuations of unpaid care. Under any replacement-cost valuation that applies one wage to all childcare hours, counting secondary childcare raises the value of women's childcare by the ratio of secondary to primary minutes: between 5% and 28% across the duration conventions. The proportional correction does not depend on the wage chosen, which makes it a useful input for national satellite accounts of household production in India.

The contribution is to the measurement of unpaid work. The paper shows, for a large lower-middle-income country with a national survey, how much the measured level of and gender gap in unpaid care depend on the treatment of secondary activities; it bounds that dependence under explicit duration conventions; and it locates secondary care in the day by matching it to the main activity it accompanies. The results bear on the reporting of SDG indicator 5.4.1, which by convention counts only primary activities, and on the design of satellite accounts.

The rest of the paper is organised as follows. Section 2 reviews the literature. Section 3 describes the data and the measures. Section 4 presents the results, Section 5 the valuation exercise, and Section 6 discusses interpretation, limitations and implications.

2 Related literature

2.1 Time intensity and overlapping activities

The idea that the intensity of work matters alongside its duration was put on the agenda of economics by Floro (1995). Hours-based measures treat an hour of cooking as the same whether or not a child is being watched at the same time. Floro (1995) argued that overlapping tasks are a dimension of work intensity, that they are concentrated among poor women who cannot buy substitutes for their time, and that they carry costs to well-being that hours do not register. Floro and Miles (2003) took the argument to data, using the Australian time-use diary, which records a secondary activity alongside the primary one, to show that overlapping activities are common and differ systematically between women and men and across household types. Floro and Pichetpongsa (2010) studied home-based workers in Thailand, many of whom combine paid production with childcare and domestic work in the same space and time, and related their work intensity to measures of well-being.

These studies raise a question that the household production framework of Becker (1965) and Gronau (1977) does not address directly: if time is an input into the production of household commodities, what is the input when one minute produces two commodities? A purely additive time budget answers by fiat, assigning the minute to one commodity. Multitasking is then invisible in the data and in the model. The approach in this paper is to measure the extent of overlap first, and to show how much the conventional answer depends on the fiat.

Diary design shapes what is recorded. Kitterød (2001) examines whether recording parallel activities in a diary changes the way people report their main activities, a question directly relevant to comparisons between surveys with and without a secondary column. Gershuny and Sullivan (1998) and Juster and Stafford (1991) discuss the strengths of diary data relative to stylised questions, including their ability to capture the sequence and simultaneity of activities. The United Nations guidance on time-use surveys (United Nations 2005) and the international classification of activities (United Nations Statistics Division 2016) both make provision for recording simultaneous activities, but leave the choice of how to use them in summary statistics to the producer. The India 2024 survey's combination of an activity list per slot, a multiple-activity flag, a simultaneity flag and a main-activity flag is richer than the single secondary column of most diaries.

2.2 Measuring childcare: primary, secondary and supervisory time

Childcare is the activity most affected by single-activity accounting, because so much of it is supervision performed alongside something else. Zick and Bryant (1996) showed early on that adding secondary childcare substantially raises measured parental time in the United States. Folbre et al. (2005) distinguished primary care, secondary care and time when a child is merely in the parent's care or present, and showed that estimates of family time devoted to children vary severalfold across these definitions. The essays in Folbre and Bittman (2004) develop the distinction between activity, proximity and responsibility as bases for measuring care. Folbre (2006) argues

that measures of care should capture supervisory responsibility, which constrains the carer's other choices even when no active care is performed.

Survey design determines which of these concepts can be measured. The American Time Use Survey does not record secondary activities in general; it measures secondary childcare through a summary question on the time a child under 13 was in the respondent's care, which Allard et al. (2007) compare with the secondary childcare columns of earlier US diaries. Economic analyses using both primary and secondary care include Kalenkoski et al. (2005) and Kalenkoski et al. (2007), who compare parents' childcare time across family structures in the United Kingdom and the United States, and Kalenkoski and Foster (2008), who distinguish primary, secondary and passive care in Australian households. Kimmel and Connelly (2007) model mothers' allocation of time among caregiving, leisure, home production and paid work, and Connelly and Kimmel (2009) model how spouses' characteristics affect each parent's non-market time. Hallberg and Klevmarken (2003) study parents' joint time allocation to children in Sweden, and Guryan et al. (2008) and Sayer et al. (2004) document the education gradient and long-run rise in parents' time with children.

The gendered character of secondary care is a recurring finding. Using Australian data, Craig (2006) shows that mothers' time with children involves more physical and routine care, more solo responsibility and more multitasking than fathers', and Craig (2007) develops the argument at book length. Craig and Bittman (2008) estimate the incremental time costs of children and show that they fall mainly on mothers. Craig and Mullan (2011) compare how mothers and fathers share childcare across countries, distinguishing types of care. The common implication is that a measure restricted to primary childcare will understate mothers' care relative to fathers' if mothers' share of secondary care exceeds their share of primary care. Whether that is true in India is an empirical question that this paper answers.

2.3 Leisure quality and fragmentation

A parallel literature argues that the quality of free time matters alongside its quantity. Bittman and Wajcman (2000) show that women's leisure is more fragmented than men's and more often combined with unpaid work or childcare, so that equal leisure minutes do not mean equal leisure. Mattingly and Bianchi (2003) find that US women have less free time than men and that their free time is more often "contaminated" by the presence of children or by other activities. Sullivan (1997) argues that women's domestic time is experienced as more fragmented and more often combined with other tasks. Sevilla et al. (2012) distinguish quantity and quality dimensions of leisure in their analysis of leisure inequality in the United States, and Aguiar and Hurst (2007) measure long-run trends in leisure quantity.

Time-poverty analysis connects these themes to welfare. Vickery (1977) proposed that poverty measures account for the time needed for household production as well as money income. Blackden and Wodon (2006) and Bardasi and Wodon (2010) apply the concept in sub-Saharan Africa, where long hours of combined paid, own-use and domestic work leave many women time poor. Hamermesh

and Lee (2007) study the subjective sense of time pressure across countries. Secondary care during free time is a natural measure of the quality dimension: it identifies leisure that is not fully discretionary because someone must be watched or something must be done.

2.4 Multitasking and well-being

Whether multitasking is costly or efficient is an open question. On the one hand, doing two things at once saves time; a parent who can mind a child while cooking produces two outputs from one hour. On the other, laboratory evidence on task switching shows that alternating between tasks carries cognitive costs (Rubinstein et al. 2001), and survey evidence links multitasking to stress. Offer and Schneider (2011) find that mothers in US dual-earner families multitask more than fathers, and that multitasking is associated with negative emotions and stress for mothers but not for fathers. Experience-sampling and day-reconstruction methods (Kahneman et al. 2004; Kahneman and Krueger 2006) allow well-being to be measured episode by episode; Musick et al. (2016) use such data to show that mothers report less positive experiences than fathers during time with children, and Connelly and Kimmel (2015) examine how mothers and fathers feel during different childcare activities. The India 2024 survey does not measure affect, so this paper cannot test the well-being consequences of multitasking directly. What it can show is who multitasks, in which combinations, and how much of the multitasking is care, which identifies the groups whose time the well-being literature suggests is most costly.

2.5 SDG 5.4.1 and the measurement of unpaid work

SDG indicator 5.4.1 measures the proportion of time spent on unpaid domestic and care work. It is usually computed from primary activities only, so that each diary sums to 24 hours. The convention is defensible: it avoids double counting and makes the indicator comparable across surveys that do and do not record secondary activities. But it means that the indicator measures the share of the day in which unpaid work is the main activity, not the share of the day in which it is performed. Reviews of time-use surveys in the global South (Esquivel et al. 2008; Esquivel 2011; Budlender 2010; Antonopoulos and Hirway 2010) stress that the treatment of simultaneous activities is one of the design choices that most affect cross-country comparability, alongside the activity classification and the choice between diaries and stylised questions. The 2013 statistical standard on work (International Labour Organization 2013) recognises own-use provision of services, which includes unpaid domestic and care services, as a form of work, and so raises the stakes of measuring it completely. Global compilations (Charmes 2019; Miranda 2011) report large gender gaps in primary-activity unpaid work, but say less about the secondary dimension.

For India, Hirway and Jose (2011) use the 1998–99 pilot survey to show how time-use data reveal women's work missed by labour force surveys, and the 2019 survey (National Statistical Office 2020) established national primary-activity estimates. The 2024 survey is the second national round.

2.6 Valuing unpaid work

The case for valuing household production goes back to Reid (1934), whose third-person criterion defines an activity as productive if it could be delegated to a paid third party. Hawrylyshyn (1977) formalised the criterion, and Goldschmidt-Clermont (1982) reviewed the valuation methods: opportunity cost, which values time at the wage the person could earn, and replacement cost, which values it at the wage of a generalist or specialist who could be hired to do the task. Waring (1988) made the political case that national accounts that ignore women's unpaid work misrepresent the economy. Ironmonger (1996) proposed gross household product, and Abraham and Mackie (2005) set out a framework for nonmarket satellite accounts in the United States, which Landefeld et al. (2009) implemented with American Time Use Survey data. Stiglitz et al. (2009) recommended that measures of economic performance include non-market household production. Applications to developing countries include Dong and An (2015) for China and Charmes (2019) for a global compilation.

Secondary activities pose a specific problem for valuation. Under the third-person criterion, minding a child while cooking is two services that could each be bought, so a replacement-cost valuation would count both. But the time input is a single hour, and valuing it twice at the full replacement wage overstates the cost of hiring substitutes if one substitute could do both tasks. One response is to exclude secondary activities from the valuation; another is to value them at a reduced rate. This paper does not take a position on the right rate. It reports the hours under alternative conventions and shows how much each adds to the valuation, in proportional terms that do not depend on the wage. It does not impute a wage, because the survey does not collect one and no external wage series is used here.

3 Data and measurement

3.1 The survey

The Time Use Survey 2024 was conducted by the National Statistics Office over January to December 2024 (National Statistics Office 2025). Every member aged six and above in sampled households reports a diary for the 24 hours from 4:00 a.m. on the day before the interview. The diary is divided into 48 half-hour slots. For each slot, the respondent reports every activity that lasted ten minutes or more, each coded to the 3-digit ICATUS 2016 classification (United Nations Statistics Division 2016). Three flags describe the slot: whether more than one activity was performed (column 6 of Block 6, coded 1 = yes, 2 = no); if so, whether any were performed simultaneously (column 7, same coding); and whether the activity was the "most important" in the slot (column 9, 1 = major, 2 = minor). The release I use has one row per activity record. Of the 10,211,478 records, 17.5% are minor. The major records partition the day: they sum to 1,440 minutes for 100.0% of diaries.

The analysis sample consists of the 388,847 diarists aged 15 and above (194,942 women and 193,905 men). Persons are identified by first-stage unit, household and person serial number. All estimates

use the survey weights, and standard errors are clustered on the first-stage unit, the primary sampling unit (Cameron and Miller 2015; Abadie et al. 2023). Gaps are differences in weighted means. Regressions are weighted least squares; for outcomes with many zeros I use linear models rather than Tobit, following Stewart (2013), because a zero on a single diary day does not imply a corner solution in the person's long-run allocation (Frazis and Stewart 2012).

3.2 Measures

Secondary activity. Secondary minutes are the sum of the recorded durations of a person's minor records, in total and by ICATUS category. The secondary categories use the same code lists as the primary categories, so that secondary childcare, for instance, covers the same activities as primary childcare. I also count the number of minor records, which does not depend on how duration is assigned.

Multi-activity slots. Minutes in multi-activity slots are the major-activity minutes in slots that the respondent flagged as containing more than one activity, and minutes in simultaneous slots those in slots flagged as containing simultaneous activities. These measures use only major-activity durations, which are well defined because the major records partition the day.

Childcare. Primary childcare is the time recorded against major childcare records (ICATUS 411–419, and accompanying own children, 442). Within it, passive childcare is "minding children" (ICATUS 416), recorded as the main activity. Inclusive childcare adds secondary childcare to primary childcare. The share of childcare that is secondary is secondary minutes divided by inclusive minutes, computed from population means.

Inclusive unpaid work. The SDG 5.4.1 numerator is primary unpaid domestic and care work (ICATUS divisions 3 and 4). The inclusive version adds secondary domestic and care minutes. Unlike the primary measure, the inclusive measure can exceed the time available for unpaid work as a main activity, because a minute may be counted twice: once as the main activity and once as the secondary activity. It is a measure of activity performed, not of time allocated.

Free time and meals. Using the start time of each record, I match each minor record to the major episode whose interval contains it (match rate: 100.0%). Free time is the major time in socialising, community and religious activities and in culture, leisure, media and sports (divisions 7 and 8). Free time shared with care is free time during which a childcare, adult-care or domestic activity was recorded as minor. I report both an indicator for any such free time, which does not depend on minor durations, and the share of free time so shared. Eating while working is an indicator for eating recorded as a minor activity during employment, own-use production, domestic work or care. Eating as a secondary activity is the share of a person's eating time (primary plus secondary) that is recorded as minor.

3.3 The duration of minor records

The release does not document how duration is assigned to minor records. A minor record in a half-hour slot might carry the full slot, the respondent's estimate of the time spent on it, or a share of the slot. Because the inclusive measures add minor durations to major durations, their level depends on this convention. I address the problem in two ways. First, several measures do not depend on it: the number of secondary records, the indicators for any secondary activity, any secondary childcare, any free time shared with care and any eating while working, and the minutes in multi-activity slots. Second, I re-estimate the inclusive childcare measure under four conventions:

1. *Recorded duration*: minor minutes as recorded.
2. *Half duration*: each minor record counts for half its recorded duration, as if the slot were divided equally between the major and the minor activity.
3. *Ten-minute floor*: each minor record counts for ten minutes, the survey's threshold for recording an activity. This is a lower bound on the time in which the minor activity was performed.
4. *Net of overlap*: recorded duration, excluding secondary childcare recorded during primary childcare, so that care time is not counted twice when one form of childcare accompanies another.

A fifth convention, counting no secondary time at all, is the primary-activity measure itself. Conventions 2 and 3 give the inclusive measure a lower weight on secondary care than convention 1 when, as is typical in slot-based diaries, recorded minor durations equal or approach the slot length.

Table 1: Secondary activities by sex, persons aged 15 and above

Outcome	Men	Women	Difference (W - M)
Secondary activity minutes	176.61 (1.76)	215.56 (2.04)	38.95***
Secondary records (number)	3.59 (0.03)	4.54 (0.04)	0.95***
Any secondary activity (share)	0.81 (0.00)	0.84 (0.00)	0.03***
Secondary childcare minutes	3.40 (0.09)	13.71 (0.25)	10.31***
Any secondary childcare (share)	0.04 (0.00)	0.12 (0.00)	0.08***
Primary childcare minutes	13.98 (0.16)	49.54 (0.37)	35.55***
Passive share of primary childcare	0.05 (0.00)	0.03 (0.00)	-0.02***
Childcare including secondary	17.39 (0.20)	63.25 (0.48)	45.86***
Secondary care and domestic minutes	10.53 (0.19)	66.38 (0.77)	55.85***
Primary unpaid work (SDG 5.4.1 numerator)	43.25 (0.32)	331.91 (0.92)	288.66***
Unpaid work including secondary	53.78 (0.42)	398.29 (1.27)	344.51***
Secondary free-time minutes	120.99 (1.35)	103.38 (1.25)	-17.60***
Eating recorded as secondary (share)	0.13 (0.00)	0.18 (0.00)	0.04***
Minutes in multi-activity slots	158.74 (1.47)	189.84 (1.65)	31.10***
Minutes in simultaneous-activity slots	73.41 (0.98)	74.97 (1.03)	1.56***
Any free time shared with care (share)	0.04 (0.00)	0.11 (0.00)	0.07***
Free time shared with care (share)	0.01 (0.00)	0.04 (0.00)	0.03***
Any eating while working (share)	0.07 (0.00)	0.28 (0.00)	0.21***

Notes: Survey-weighted means; FSU-clustered standard errors in parentheses. Difference is women minus men; * p<0.10, ** p<0.05, *** p<0.01. Minutes are per diary day. Shares of free time and eating are among persons with any free time or eating.

4 Results

4.1 How much activity is secondary, and who does it

Table 1 reports the measures by sex. Women record 216 minutes of secondary activity a day and men 177, a raw gap of 39.0 minutes (SE 1.1). Women are 3.3 percentage points (SE 0.2) more likely than men to record any secondary activity, and they record 0.9 records (SE 0.0) more secondary records a day. Because the number of records does not depend on the duration convention, the direction of the gap in secondary activity does not depend on how minor records are timed. On the slot-based measure, women spend 31.1 minutes (SE 0.9) more major-activity time in slots they flag as containing more than one activity. Multitasking is thus more common among women on every measure the survey offers.

Care and domestic work account for 31% of women's secondary minutes and 6% of men's: women record 66 minutes a day of secondary care and domestic work, men 11. Secondary childcare alone is 14 minutes a day for women and 3 for men; 12.5% of women and 4.4% of men record any. Primary childcare also contains a passive component, "minding children" recorded as the main activity; among persons with any primary childcare, it accounts on average for 3.4% of women's primary childcare and 5.4% of men's.

Table 2: Gender gaps in secondary activity, conditional on characteristics

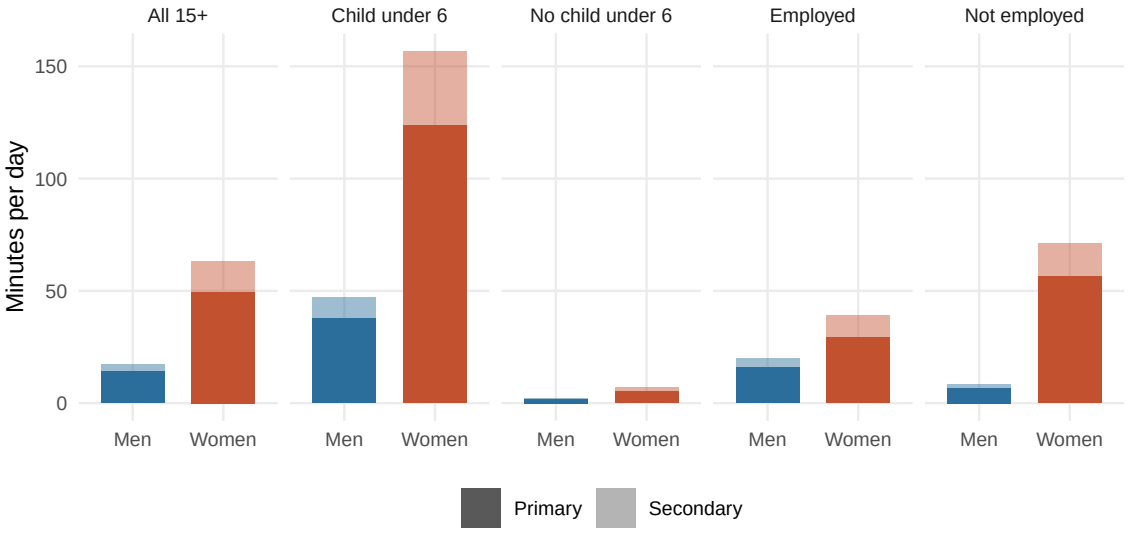
	Sec. min. (1)	Any sec. CC (2)	Sec. CC (3)	Incl. CC (4)	Incl. unpaid (5)	Any FT w. care (6)
Woman	37.3***	0.064***	7.44***	31.0***	275.8***	0.059***
	(1.17)	(0.001)	(0.179)	(0.341)	(1.16)	(0.002)
Observations	388,847	388,847	388,847	388,847	388,847	388,847
R^2	0.09661	0.13888	0.10044	0.40747	0.61775	0.04982
State fixed effects	✓	✓	✓	✓	✓	✓
Day of week fixed effects	✓	✓	✓	✓	✓	✓

Notes: Persons 15+. Each column is a weighted least squares regression of the outcome on an indicator for women. Outcomes: (1) secondary activity minutes; (2) any secondary childcare; (3) secondary childcare minutes; (4) childcare including secondary; (5) unpaid domestic and care work including secondary; (6) any free time shared with care or domestic work. Controls: age band, education, marital status, urban, consumption quintile, household size, numbers of members under 6, aged 6–14 and 60+, employment status, proxy report; state and day-of-week fixed effects. FSU-clustered SEs in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table 2 shows the gaps conditional on individual and household characteristics. The conditional gap in secondary minutes is 37.3 minutes (SE 1.2), in the probability of any secondary childcare 6.4 percentage points (SE 0.1) higher for women, and in secondary childcare minutes 7.4 minutes (SE 0.2). Women's excess secondary childcare is therefore not explained by differences in age, education, marital status, household composition, employment or consumption. The household fixed-effects estimates in the appendix (Table 8) compare women and men in the same household, who share the

same children, dwelling, technology and consumption. The within-household gap is 37.7 minutes (SE 0.9) in secondary minutes and 7.1 minutes (SE 0.2) in secondary childcare. The gender gap in secondary care thus persists between women and men in the same household; it is not an artefact of differences between the households in which they live.

4.2 What counting secondary care does to the childcare gap



Persons 15+. Survey−weighted means. Solid: primary childcare; light: secondary childcare at recorded duration.

Figure 1: Primary and secondary childcare by sex, household type and employment status

Figure 1 shows primary and secondary childcare by sex, for all persons and by household type and employment status. Among women, secondary care is 22% of total childcare; among men, 20%. The women-to-men ratio is 3.5 on the primary measure and 3.6 on the inclusive measure. Because women's share of secondary childcare exceeds their share of primary childcare, the primary-activity measure understates the gender imbalance in childcare. In minutes, the gap in inclusive childcare is 45.9 minutes (SE 0.4), against 35.6 minutes (SE 0.3) on the primary measure.

Table 3: Childcare under alternative conventions for the duration of secondary records

Measure	Women	Men	Gap (SE)	Ratio W/M	Secondary share, women	Hours a year, women
Primary activity only (SDG convention)	49.5	14.0	35.6*** (0.3)	3.54	0.0%	301
(1) Primary + secondary at recorded duration	63.2	17.4	45.9*** (0.4)	3.64	21.7%	385
(2) Primary + secondary at half duration	56.4	15.7	40.7*** (0.3)	3.60	12.2%	343
(3) Primary + 10 minutes per secondary record	51.9	14.6	37.3*** (0.3)	3.56	4.5%	316
(4) As (1), excluding secondary care during primary care	62.4	17.2	45.3*** (0.4)	3.64	20.7%	380

Notes: Persons 15+. Minutes per day, survey-weighted. Gap is women minus men with FSU-clustered SE; * p<0.10, ** p<0.05, *** p<0.01. Secondary share is the share of women's childcare under each convention that is secondary. Hours a year are daily minutes times 365/60.

Table 3 reports the inclusive measure under the four duration conventions. The widening of the gender gap from counting secondary care equals the gap in the secondary component itself, so its sign and significance can be read from the secondary gap under each convention: 10.3 minutes (SE 0.2) at

recorded duration, half that at half duration, 1.8 minutes (SE 0.0) under the ten-minute floor, and 9.7 minutes (SE 0.2) net of overlap with primary care. Under every convention, counting secondary care widens the gender gap in childcare by a statistically significant amount. The women-to-men ratio on the inclusive measures ranges from 3.6 to 3.6, against 3.5 on the primary measure. The share of women's childcare that is secondary ranges from 12% under half duration to 22% at recorded duration.

The half-duration convention is a useful benchmark because it divides the slot between the major and the minor activity. Even on this convention, women's secondary childcare adds 7 minutes a day, or 42 hours a year, to their measured childcare. On the ten-minute floor, which assumes that each recorded episode of secondary care lasted no longer than the survey's recording threshold, it adds 2 minutes a day.

4.3 Where secondary care happens

Table 4: What women and men are doing while providing secondary childcare

Main activity during secondary childcare	Women	Men
Unpaid domestic work	34.6%	2.0%
Culture, leisure, media, sports	22.0%	49.6%
Eating	16.4%	14.4%
Socialising, community, religion	12.3%	16.3%
Childcare	6.0%	6.7%
Hygiene	3.7%	6.6%
Sleep	1.7%	0.5%
Employment	1.7%	2.0%
Own-use goods production	0.9%	0.4%
Learning	0.3%	0.3%
Other care	0.2%	0.3%
Other self-care	0.2%	0.5%
Volunteering	0.0%	0.0%
Dependent-adult care	0.0%	0.1%
Receiving care	0.0%	0.0%
Care of non-dependent adults	0.0%	0.1%

Notes: Persons 15+. Share of each sex's weighted secondary childcare minutes, by the main activity of the episode containing the secondary record. – indicates no secondary childcare recorded by that sex.

Table 4 shows what women and men are doing when they record childcare as a secondary activity. For women, the most common main activity is unpaid domestic work, which accounts for 35% of their secondary childcare minutes. Secondary childcare is thus mainly a matter of combining care with other work, the pattern that the literature on work intensity emphasises. This matters for interpretation. Secondary care during domestic work is overlapping work: two services from one hour. Secondary care during free time or self-care is care that cuts into time conventionally classed as non-work. Secondary care during paid work is the combination that Floro and Pichetpongsa (2010) emphasise for home-based workers.

Table 5: Most common combinations of main and secondary activity

Sex	Main activity	Secondary activity	Share of secondary minutes
Men	Eating	Socialising, community, religion	20.4%
Men	Culture, leisure, media, sports	Socialising, community, religion	10.4%
Men	Employment	Socialising, community, religion	8.9%
Men	Eating	Culture, leisure, media, sports	7.2%
Men	Culture, leisure, media, sports	Culture, leisure, media, sports	4.0%
Men	Socialising, community, religion	Eating	3.6%
Women	Unpaid domestic work	Unpaid domestic work	13.0%
Women	Eating	Socialising, community, religion	12.5%
Women	Unpaid domestic work	Socialising, community, religion	8.1%
Women	Unpaid domestic work	Eating	6.6%
Women	Culture, leisure, media, sports	Socialising, community, religion	6.1%
Women	Eating	Unpaid domestic work	5.6%

Notes: Persons 15+. Share of each sex's weighted secondary minutes.

Table 5 widens the view to all secondary activities, listing the six most common combinations of main and secondary activity for each sex. It shows whether women's and men's secondary activity consists of the same combinations in different amounts or of different combinations altogether.

4.4 Who does the hidden care: young children and employment

Table 6: Secondary childcare by household type and employment status

Group	Women: sec.	Men: sec.	Gap in sec. (SE)	Sec. share, women	Ratio: primary	Ratio: incl. sec.	N
All persons 15+	13.7	3.4	10.3*** (0.2)	21.7%	3.54	3.64	388,847
Child under 6 in household	33.3	9.2	24.1*** (0.5)	21.2%	3.27	3.34	141,049
No child under 6 in household	2.0	0.4	1.6*** (0.1)	27.4%	2.88	3.22	247,798
Employed (usual status)	10.1	3.9	6.2*** (0.2)	25.7%	1.82	1.97	198,969
Not employed (usual status)	14.9	1.6	13.4*** (0.3)	20.9%	8.27	8.52	189,878

Notes: Persons 15+. Minutes per day, survey-weighted. Gap is women minus men with FSU-clustered SE; * p<0.10, ** p<0.05, *** p<0.01. Secondary share: secondary as a share of women's childcare including secondary. Ratios are women's to men's mean childcare. A child under 6 is any household member under 6 (household size minus diarists); employment is usual principal activity status.

One would expect women's secondary care to be concentrated where the demand for supervision is greatest and where women's time is most constrained. Table 6 and Figure 1 report the measures separately for persons living in households with and without a child under six, and for persons who are and are not employed in their usual principal status. Children under six do not keep diaries; I identify their presence as household size minus the number of diarists, which is valid if every member aged six and above has a diary (method in this release: direct count (file contains members under 6)). The comparison is between household types, not between parents and non-parents: in extended households, a child under six may be a grandchild, niece or nephew.

In households with a child under six, the gender gap in secondary childcare is 24.1 minutes (SE 0.5); in households without one, 1.6 minutes (SE 0.1). Women in households with a young child record 33 minutes of secondary childcare a day, and secondary care makes up 21% of their childcare. In these households the women-to-men childcare ratio moves from 3.3 on the primary measure to 3.3

on the inclusive measure.

Among the employed, the gap in secondary childcare is 6.2 minutes (SE 0.2); among those not employed, 13.4 minutes (SE 0.3). Employed women record 10 minutes of secondary childcare a day and women who are not employed 15, a difference (employed minus not employed) of -4.8 minutes (SE 0.3). Secondary care is 26% of employed women's childcare and 21% of non-employed women's. Employed women thus do a larger share of their childcare as a secondary activity, consistent with care being fitted around paid work.

Table 7: Heterogeneity in the gender gap in childcare

	Sec. CC (1)	Incl. CC (2)	Sec. CC (3)	Incl. CC (4)
Woman	0.585***	-0.702***	6.82***	32.8***
	(0.113)	(0.209)	(0.202)	(0.407)
Child under 6	8.16***	38.7***		
	(0.284)	(0.542)		
Employed	-2.22***	-14.0***	-4.44***	-19.5***
	(0.204)	(0.356)	(0.167)	(0.365)
Woman × Child under 6	21.5***	99.5***		
	(0.466)	(0.746)		
Woman × Employed			1.14***	-3.38***
			(0.303)	(0.557)
Observations	388,847	388,847	388,847	388,847
R^2	0.12795	0.50194	0.10048	0.40753
State fixed effects	✓	✓	✓	✓
Day of week fixed effects	✓	✓	✓	✓

Notes: Persons 15+. Weighted least squares. Outcomes: secondary childcare minutes (columns 1 and 3) and childcare including secondary (columns 2 and 4). Columns 1–2 interact the indicator for women with an indicator for a child under 6 in the household and control for employment; columns 3–4 interact it with usual-status employment and control for the number of children under 6. Other controls as in Table 2; state and day-of-week fixed effects. FSU-clustered SEs. * p<0.10, ** p<0.05, *** p<0.01.

Table 7 tests these differences with interactions, holding the other characteristics constant. The interaction of the indicator for women with the presence of a child under six is 21.5 minutes (SE 0.5) for secondary childcare and 99.5 minutes (SE 0.7) for inclusive childcare, so the gender gap in secondary childcare is larger in households with a child under six. The interaction with employment is 1.1 minutes (SE 0.3) for secondary childcare and -3.4 minutes (SE 0.6) for inclusive childcare: the gap is larger among the employed than among those not employed. Employment does not relieve women of secondary care; if anything, the gap is larger among the employed, consistent with a double burden in which supervision is fitted around paid work.

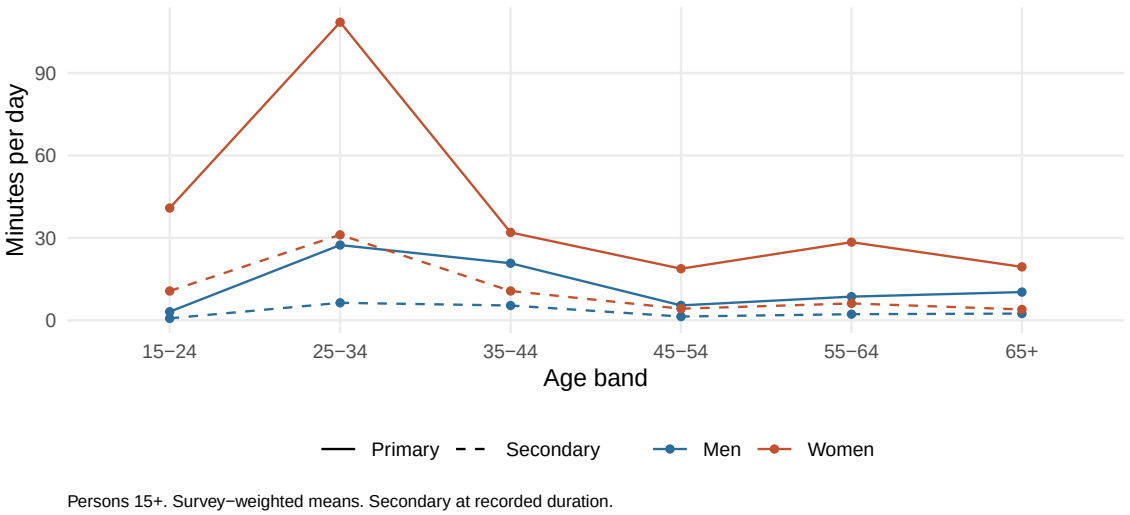


Figure 2: Primary and secondary childcare by age and sex

Figure 2 shows primary and secondary childcare over the life course. The age profile of women's secondary care shows whether hidden care is concentrated in the prime childbearing years or extends to older women, who in extended households may supervise grandchildren while doing other work.

4.5 Leisure quality and meals

The difference between women and men in the probability of having any free time shared with care or domestic work is 7.3 percentage points (SE 0.1). Among persons with free time, the difference in the share of free time so shared is 2.6 percentage points (SE 0.1), and the conditional difference in the probability (Table 2) is 5.9 percentage points (SE 0.2). Measured free time therefore overstates women's leisure relative to men's, the pattern that Bittman and Wajcman (2000) and Mattingly and Bianchi (2003) found in rich countries.

For meals, the difference between women and men in the probability of eating while working, doing domestic work or caring is 20.6 percentage points (SE 0.3). The difference in the share of eating time recorded as a secondary activity is 4.3 percentage points (SE 0.1). Women's meals are more often an accompaniment to other activity than the main activity of the slot. Eating as a secondary activity is a marker of time pressure rather than a burden in itself; it identifies people for whom a meal is not a break.

4.6 An inclusive SDG 5.4.1 measure

On the primary-activity convention used for SDG indicator 5.4.1, women do 332 minutes of unpaid domestic and care work a day and men 43, a ratio of 7.7. Adding secondary domestic and care activity raises these to 398 and 54 minutes, a ratio of 7.4. The gap rises from 288.7 minutes (SE 0.9) to 344.5 minutes (SE 1.2), and the conditional gap on the inclusive measure is 275.8 minutes (SE 1.2). The primary-activity indicator therefore understates the gender gap in unpaid work as well as

its level. Because a minute can be counted twice on the inclusive measure, it should be reported alongside the official indicator, not in place of it.

4.7 Proxy reporting

About 27% of women's diaries and 40% of men's are reported by another household member. Proxy informants may not know about, or may not think to report, the secondary activities of others. Conditional on demographics, household composition and sex, the coefficient on proxy reporting in a regression of secondary minutes is -12.5 minutes (SE 2.1), and the difference between women and men in this proxy effect is -8.8 minutes (SE 2.4). Proxy-reported diaries thus record less secondary activity than self-reported ones. The proxy effect is more negative for women's diaries than for men's, so proxy reporting may understate women's secondary activity in particular. Because men's diaries are more often proxy-reported, under-reporting by proxies would inflate the raw gender gap. Restricting the sample to self-reported diaries, the gap is 42.9 minutes (SE 1.4) in secondary minutes and 11.6 minutes (SE 0.2) in secondary childcare. The main conclusion about secondary childcare is the same in self-reported diaries.

5 Valuing hidden care

The survey does not collect wages, and I do not import an external wage series. A monetary valuation of secondary care is therefore outside the scope of this paper. What the data do determine is the quantity side of any valuation, and, for the common replacement-cost approach, the proportional effect of counting secondary care on the value of childcare.

Let h^P and h^S be a group's mean primary and secondary childcare hours, w the replacement wage, and $\lambda \in [0, 1]$ the weight placed on secondary hours. A single-wage replacement-cost valuation is $V(\lambda) = w(h^P + \lambda h^S)$. The proportional uplift from counting secondary care is

$$\frac{V(\lambda) - V(0)}{V(0)} = \lambda \frac{h^S}{h^P},$$

which does not depend on w. The choice of λ embodies the valuation debate discussed in Section 2: $\lambda = 1$ treats minding a child while cooking as two fully separate services, as the third-person criterion suggests (Reid 1934; Hawrylyshyn 1977); $\lambda = 0$ is the primary-activity convention; and intermediate values reflect the view that one hired worker could supply both services in part. The duration conventions in Table 3 correspond to different values of λh^S.

For women aged 15 and above, secondary childcare at recorded duration is 83 hours a year per woman, against 301 hours of primary childcare. With $\lambda = 1$ the uplift is 28%; at half duration, 14%; under the ten-minute floor, 5%; and net of overlap with primary care, 26%. For men, the uplift at recorded duration is 24%. The women-minus-men gap in secondary childcare is 63 hours a year per person. These proportions can be applied directly to any existing replacement-cost estimate of the

value of childcare in India that was computed from primary activities.

At the population level, the survey weights for women aged 15 and above sum to 423.3 million. If the weights are population multipliers, women's secondary childcare amounts to 35.31 billion hours a year, and their secondary care and domestic work together to 170.92 billion hours. A valuation would multiply these hours by a wage for childcare workers. The choice between a generalist wage, a specialist wage and an opportunity-cost wage (Goldschmidt-Clermont 1982; Abraham and Mackie 2005) would matter for the level, but not for the proportional uplift reported above.

Two cautions apply. First, valuing secondary care at the full replacement wage assumes that a hired substitute would have been needed for the supervision. For care during domestic work, a single hired worker might supply both services, which argues for $\lambda < 1$. Second, the opportunity-cost method values an hour at the carer's own wage, and so assigns the same value to an hour regardless of how many activities it contains; under that method, secondary care adds nothing to the valuation unless it displaces another activity. Across these methods, the replacement-cost uplift with $\lambda = 1$ is an upper bound on what counting secondary care adds to the value of household production, and the half-duration and ten-minute-floor conventions give more conservative figures.

6 Discussion

6.1 Interpretation

Single-activity accounting misses a substantial part of what Indian women do. Counting secondary childcare raises the measured women-to-men ratio in childcare, and the widening of the gap holds under each convention for the duration of secondary records. Women's free time is more often shared with care or domestic work than men's. Women more often eat while working. These are the dimensions of time quality that Floro (1995) and Bittman and Wajcman (2000) argued are missing from hours-based measures.

Three interpretations of secondary care are possible, and the data cannot fully distinguish them. The first is efficiency: combining tasks saves time, so a household that relies on secondary care produces more from the same hours. The second is constraint: secondary care is supervision that must be provided because no one else is available, and it is costly because it limits what else can be done and adds to the intensity of the day (Floro 1995; Offer and Schneider 2011). The third is reporting: what one respondent records as a secondary activity, another might not record at all, so differences in secondary activity partly reflect differences in how people describe their day (Kitterød 2001). The concentration of the gap in households with young children, and the fact that secondary childcare is mostly recorded during other work (Table 4), favour reading most secondary childcare as supervision combined with other work rather than as a reporting artefact. The proxy-reporting analysis addresses part of the reporting interpretation.

The heterogeneity results bear on the policy debate about care infrastructure. That the gap is

larger among the employed suggests that paid work does not displace secondary care, consistent with care being fitted around employment. Because employment status is the usual principal status rather than work on the diary day, and because employment is a choice correlated with household circumstances, these patterns are descriptive and not causal.

6.2 Limitations

The analysis has five main limitations. First, the duration of minor records is undocumented. I address this by reporting measures that do not depend on duration and by bounding the inclusive measure under four conventions, but the levels of secondary minutes should be read with that uncertainty in mind. Second, the survey records an activity only if it lasted ten minutes or more in the slot. Brief acts of supervision, which may be the most common form of secondary care, are not recorded, so the inclusive measures are lower bounds on the time during which a person is responsible for a child. Third, the survey does not record responsibility or on-call time as such; the "in your care" concept of Allard et al. (2007) and the responsibility measures discussed by Folbre (2006) are wider than secondary childcare. Fourth, each person reports a single diary day. Gaps in means are unbiased for gaps in long-run time use, but individual-level variation overstates the dispersion of usual behaviour (Frazis and Stewart 2012). Fifth, the survey does not measure affect or stress, so the well-being costs of multitasking documented by Offer and Schneider (2011) and Musick et al. (2016) cannot be tested here.

Identification of children under six from household size minus diarists is valid only if all members aged six and above keep diaries. The validation output reports the share of households where this residual is negative, which would indicate a violation. Finally, the matching of minor records to major episodes uses start times; where a minor record begins in one major episode and continues into another, it is attributed to the first.

6.3 Implications for measurement

For SDG reporting, the results support publishing, alongside indicator 5.4.1, an inclusive measure that counts secondary unpaid care and domestic work, together with the duration convention used. The 2024 survey already collects what is needed. For national accounts, the proportional uplifts in Section 5 can be applied to any primary-activity replacement-cost valuation, and the choice of λ can be made explicit rather than set implicitly to zero. For survey design, the sensitivity of the inclusive measure to duration conventions argues for documenting how duration is assigned to minor records, and for recording supervisory responsibility directly, as the American Time Use Survey does with its "in your care" question. For policy, the 3R framework of recognising, reducing and redistributing unpaid care (Elson 2017) begins with recognition, and recognition requires that care performed alongside other activities is counted.

7 Conclusion

India's Time Use Survey 2024 records every activity performed in each half-hour slot, not just the main one. Using every record, this paper measures how much of women's and men's activity is secondary, how much of it is care, and what counting it does to the measured gender gap. Counting secondary childcare raises the women-to-men ratio in childcare, from 3.5 to 3.6 at recorded duration and to between 3.6 and 3.6 across the duration conventions. The gender gap in secondary childcare is larger in households with a child under six and larger among the employed than among those not employed. Under a single-wage replacement-cost valuation, counting secondary childcare raises the value of women's childcare by between 5% and 28%. Statistics that count one activity at a time are a convention, not a description of how time is used; for women's care work in India, the convention determines how much of that work is seen.

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Appendix

Table 8: Within-household gender gaps: household fixed effects

	Sec. min. (1)	Any sec. CC (2)	Sec. CC (3)	Incl. CC (4)	Incl. unpaid (5)
Woman	37.7***	0.060***	7.12***	29.4***	288.3***
	(0.936)	(0.001)	(0.178)	(0.347)	(1.20)
Observations	371,522	371,522	371,522	371,522	371,522
R^2	0.77877	0.58412	0.55868	0.66671	0.76296
Household fixed effects	✓	✓	✓	✓	✓

Notes: Persons 15+. Outcomes as in columns 1–5 of Table 2. Weighted least squares with household fixed effects, so the coefficient compares women and men in the same household. Singleton households are dropped. Controls: age band, education, marital status, employment status, proxy report. FSU-clustered SEs. * p<0.10, ** p<0.05, *** p<0.01.

Table 9: Gender gaps in duration-free and quality measures

	Multi slots (1)	Sec. records (2)	Eat. sec. share (3)	FT care share (4)	Eat while work (5)
Woman	30.1***	0.993***	0.041***	0.023***	0.206***
	(0.948)	(0.022)	(0.002)	(0.0006)	(0.003)
Observations	388,847	388,847	388,716	380,371	388,847
R^2	0.10537	0.19842	0.04262	0.04788	0.12008
State fixed effects	✓	✓	✓	✓	✓
Day of week fixed effects	✓	✓	✓	✓	✓

Notes: Persons 15+. Outcomes: minutes of major activity in multi-activity slots; number of secondary records; share of eating recorded as secondary; share of free time shared with care or domestic work; any eating while working (the last two only when slot pairing is available). Controls and fixed effects as in Table 2. Shares are among persons with any eating or any free time. FSU-clustered SEs. * p<0.10, ** p<0.05, *** p<0.01.

Table 10: Construction and validation of the person-day panel

Check	Value
Activity records	10,211,478
Person-days (diaries)	450,457
Households	139,487
First Stage Units	9,969
Diaries summing to 1,440 minutes	100.0%
Records flagged minor (secondary)	17.5%
Persons with varying weight	0.00%
Invalid activity codes	None
Married persons with linked spouse	88.5%
Under-6 count method	direct count (file contains members under 6)

Notes: All activity records are used; no row sampling.